W13 · EMNLP 2026 WORKSHOPS
ArabicNLP 2026 · Shared Tasks
The Fourth Arabic Natural Language Processing Conference

› ALEXIS AT DIALECTSENTEVAL SUBTASK 2

SHARED TASK DialectSentEval 2026 SUBTASK 2

Retrieval-Augmented, Preference-Tuned Generation for Arabic Sentiment Swap

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1st OF 15 SUBMISSIONS

0.9581 Sentiment Style Accuracy — official test set, rank 1 / 15

+17.3 pts SSA ahead of 2nd place — 0.9581 vs 0.7851

56.91 / 72.18 BLEU / chrF — vs. human rewrites

+0.155 SSA from one fix — a dedicated source-polarity classifier

01 › THE TASK

Flip the polarity, keep everything else

Rewrite a sentence so it carries the **opposite polarity** while its topic stays the same. Push too hard on polarity and the rewrite drifts off topic; push too hard on similarity and the polarity does not flip.

SOURCE تجربة جميلة سعر كتير مناسب شكرا لكم
'a nice experience, reasonable price, thank you'

▼ SWAP ▼

REWRITE تجربة سيئة سعر كتير غير مناسب لا شكرا عليكم
'a bad experience, unreasonable price, no thanks'

A training pair from MA'AKS: a few words change; structure and length do not.

MA'AKS (MSA PAIRS)	TRAIN	VAL	TEST
Pairs	6,752	964	1,931
Unique sources	5,517	–	–
Has source_polarity	**Yes**	**Yes**	**No**
Exact dup. of a train source	–	20.4%	22.8%

Scoring: SSA, the fraction of outputs a fixed CAMeLBERT sentiment classifier assigns to the target polarity, plus **BLEU** and **chrF** against the human rewrite.

02 › CORPUS ANALYSIS

Three data properties, three components

OBSERVATION	DESIGN RESPONSE
91.0% of training pairs within two words of the source's length; char. similarity median 0.82 (IQR 0.75–0.88)	→ **Local edit, not paraphrase** edit-tagging pool member and a chrF content term in the reranker
22.8% of test sentences byte-identical to a train or validation source	→ **Retrieval index** a gold rewrite already exists, a strong candidate at no modelling cost
323 aligned emoji flips; antonyms such as جميلة → كريهة (115×)	→ **Polarity-flip lexicons** a deterministic safety net for markers the models leave behind

Register is noisy social-media text (emoji in 60.7–63.7% of sentences); training sources lean slightly negative (3,731 vs 3,021).

03 › PRIMARY GENERATOR

Details that mattered for mT5-base

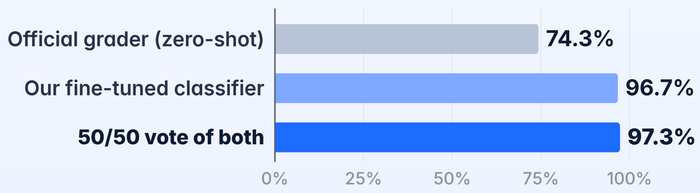
DIRECTION (VALIDATION)	CHRF	SSA
Negative → positive	78.94	94.37%
Positive → negative	84.27	96.06%

- **Asymmetric decoding:** negative-to-positive is the harder direction, so we decode with 13 beams instead of 10 when the source is predicted negative.
- **Select checkpoints by beam-search chrF**, not teacher-forced loss: the lowest-loss checkpoint wrote measurably worse text once it had to decode freely.
- Two Arabic-specific T5 checkpoints were tried and discarded (an unparseable tokenizer format; incoherent zero-shot output).

04 › THE HIDDEN CATCH

The test set omits the source polarity

The task description says source polarity is given, and train/validation carry a `source_polarity` column, but **the test file does not**. The obvious substitute, the classifier that grades the task, is right only **74.3%** of the time: roughly a quarter of the test set was being rewritten toward the wrong target, and no generator tuning could see it.



Source-polarity accuracy against gold labels. A classifier fine-tuned on the 7,716 labeled pairs reaches 96.7%; the two fail in different places, so a 50/50 vote reaches 97.3%.

≈ 1 / 4 of the test set rewritten toward the wrong target with the grader's own labels

+0.155 SSA from the dedicated classifier, the largest single gain (0.7887 → 0.9436)

05 › SYSTEM

Detect, pool, rerank, repair

1 Source-polarity detection
Dedicated + official classifier vote (97.3%) fixes the target polarity *t*, the opposite label, that everything downstream aims at.

↓

2 Candidate pool: four generators + retrieval

mT5-base	mT5-large	NLLB-200	Edit tagger	Retrieval
10 beams (13 if source negative)	6 beams	3 beams, Arabic→Arabic	KEEP / DELETE / REPLACE	gold rewrite, cosine ≥ 0.90

The primary mT5-base is fully fine-tuned on all 7,716 pairs and refined with DPO. The tagger cannot paraphrase at all, which is why it helps, and still reaches the reference for 90.6% of validation sources.

↓

3 Rerank every candidate, then repair
Retrieved rewrites must win on the same terms as generated ones. A deterministic pass then flips any emoji or antonym that still carries the source polarity.

$$s(y) = \alpha \cdot p_t(y) + (1 - \alpha) \cdot \mathrm{chrF}(y, x)$$

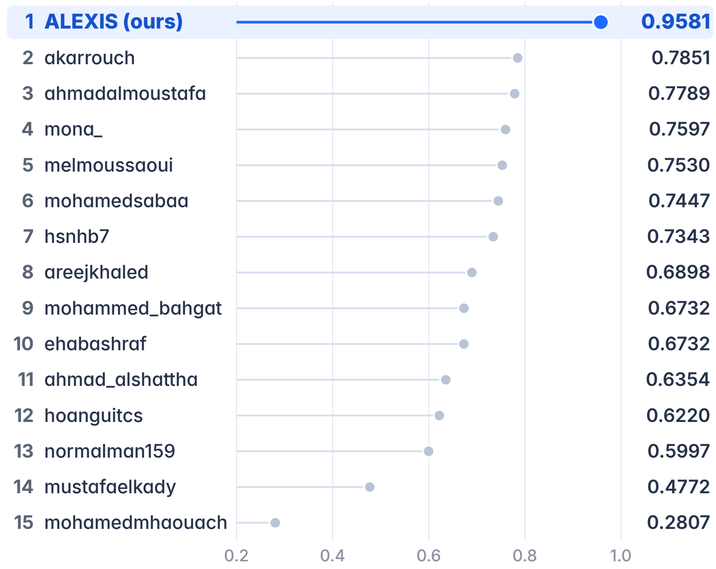
α = 0.85 when reranking · p_t: the grading classifier's raw three-way softmax probability of the target polarity · the same form of score is the reranker and the DPO reward

PREFERENCE OPTIMIZATION
DPO on 4,500 examples, 7 epochs: sample 4 candidates, add the gold rewrite, score all five with s(y). An early reward dropped neutral and renormalized two classes, so a rewrite could look on-target while the grader still read it as neutral. **Keeping the raw three-way probability closes the loophole.**

COST-AWARE DEPLOYMENT
The generation-side additions are worth about **+0.044** SSA in total, against **+0.155** for the polarity classifier alone: a latency-sensitive deployment could keep only mT5-base and the classifier.

06 › OFFICIAL RESULTS

1st of 15, 17.3 SSA points clear

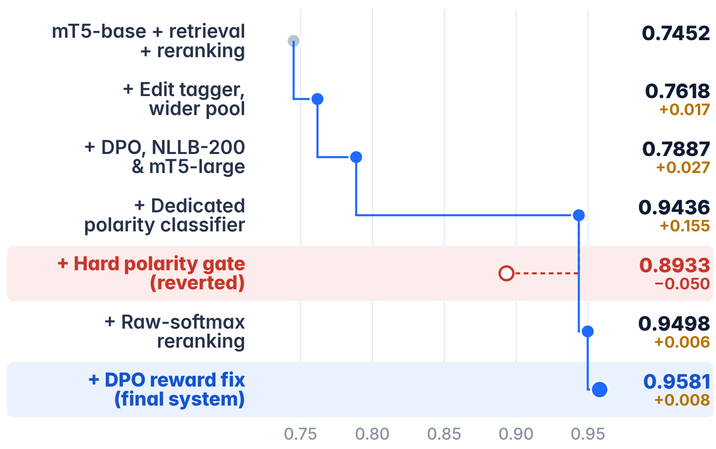


Official Subtask 2 leaderboard (SSA). The margin is specific to SSA: several systems are within a few points on BLEU and chrF and differ mainly in whether the polarity actually flipped.

#	TEAM	SSA	BLEU	CHRF
1	**ALEXIS (ours)**	**0.9581**	**56.91**	**72.18**
2	akarrouch	0.7851	49.82	70.12
3	ahmadalmoustafa	0.7789	41.48	64.45
4	mona_	0.7597	51.58	71.02

07 › ABLATION

Where the gains came from



Cumulative ablation (SSA), every row an independent leaderboard submission. The hard gate was reverted, so the next row is measured from 0.9436 and each later row from the previous retained one.

+0.044 generation-side additions: tagger, wider pool, DPO, NLLB-200, mT5-large

+0.155 the dedicated source-polarity classifier alone

A CHANGE THAT LOOKED SAFE AND WAS NOT
A hard gate discarding candidates the classifier did not label with the target polarity appeared to cost nothing on validation, then **−0.050** on the leaderboard: the same classifier chose and judged the candidates. A rule validated only against the signal it optimizes will always look correct.

VALIDATION CAN MISLEAD
Grid search on validation preferred α = 1, all weight on polarity and none on content. Submitted, it lowered BLEU and chrF on the leaderboard, so we moved only part of the way toward each validation optimum (α = 0.85).

TAKE-HOME MESSAGE

The largest single gain came from **predicting a label the task assumed would be given**. When a shared task is built around an auxiliary classifier, **measure that classifier on the sub-task it is asked to perform before trusting it**. Treat the swap as constrained editing, and let one score balance polarity and content.

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CODE github.com/Momen-Ibrahim-Fawzy/ALEXIS-DialectSentEval-Subtask2-SentimentSwap-

KEY REFERENCES
Ezzini et al. (2026). DialectSentEval 2026: Arabic Dialect Sentiment Analysis and Swapping Shared Task. ArabicNLP. · Mughaus et al. (2026). MA'AKS: Manually-curated parallel dataset for Arabic text sentiment swap. LRE. · Xue et al. (2021) mT5 · Rafailov et al. (2023) DPO · Malmi et al. (2019) LaserTagger · Mallinson et al. (2020) FELIX · Li et al. (2018) Delete, retrieve, generate · Inoue et al. (2021) CAMeLBERT · NLLB Team (2022) · Gao et al. (2023) Reward over-optimization.

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